024514

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

MATHEMATICS — 4008/2
PAPER 2

NOVEMBER 2005 SESSION 2 hours 30 minutes

Additional materials:
- Answer paper
- Geometrical instruments
- Graph paper (3 sheets)
- Mathematical tables
- Plain paper (1 sheet)

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions in Section **A** and any **three** questions from Section **B**.

Write your answers on the separate answer paper provided.
If you use more than one sheet of paper, fasten the sheets together.

Electronic calculators must not be used.

All working must be clearly shown. It should be done on the same sheet as the rest of the answer.
Omission of essential working will result in loss of marks.
If the degree of accuracy is not specified in the question and if the answer is not exact, the answer should be given to three significant figures. Answers in degrees should be given to one decimal place.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.
Mathematical tables may be used to evaluate explicit numerical expressions.

This question paper consists of 12 printed pages.

[Turn over

Section A [64 marks]

Answer **all** *the questions in this section.*

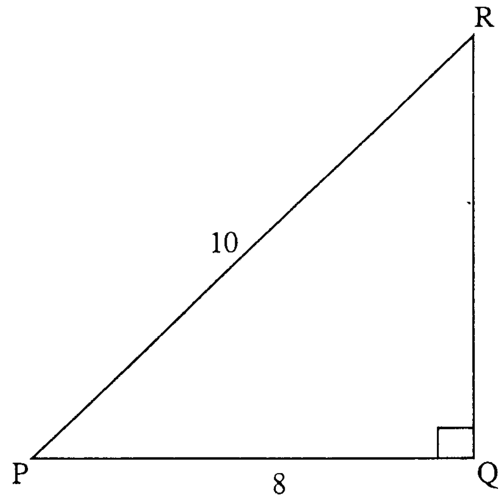
1 **(a)** Linda and Peter shared $63 000 in the ratio 2: 7 respectively. Calculate

(i) Peter's share,

(ii) the difference between their shares. [4]

(b) Subtract 6 weeks 5 days from 11 weeks 2 days, giving your answer in weeks and days. [2]

(c)



In the diagram, PQR is a right angled triangle. PQ = 8 cm, PR = 10 cm and $P\hat{Q}R = 90°$. Calculate the area of triangle PQR, giving your answer in square metres. [4]

2 **(a)** Factorise completely

(i) $ab^2 - bc$,

(ii) $6x^2 + 25x - 9$. [4]

(b) Express $\dfrac{1}{m} - \dfrac{2}{m+3}$ as a single fraction in its lowest terms. [2]

(c) Solve the equation $3(7 - x) = 6 + 2x$. [2]

(d) Given that $y = \dfrac{3x}{x - t}$,

(i) calculate the value of y when $x = 4$ and $t = 1{,}5$,

(ii) express x in terms of y and t. [5]

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3 **(a)** **(i)** Write down $1 \times 2^4 + 1 \times 2^2 + 1 \times 2^1 + 1$ as a number in base 2.

(ii) If $103_x = 67_{10}$, find the value of x. [5]

(b) Four identical balls are placed in a box. The balls are numbered 2; 3; 4 and 5. Two balls are picked from the box at random without replacement and the total score is recorded. Find the probability that

(i) the total score is 7,

(ii) the product of the scores is 6. [5]

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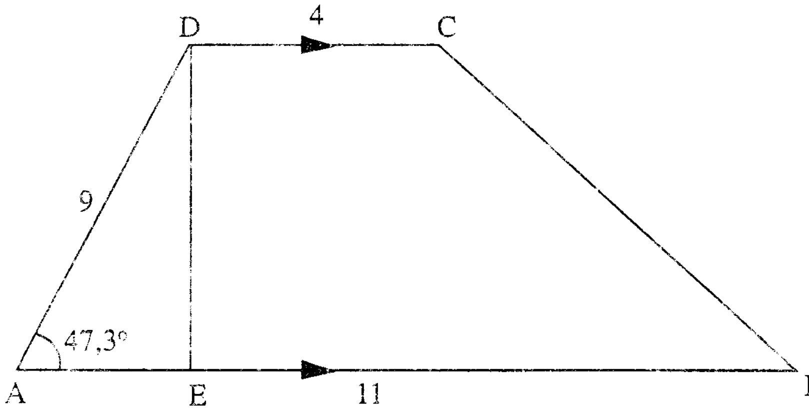
4 (a)



In the diagram, ABCD is a trapezium in which AB is parallel to DC, AB = 11 cm, DC = 4 cm, AD = 9 cm and $B\hat{A}D = 47{,}3°$. E is a point on AB such that DE is perpendicular to AB.

Calculate

(i) the length of DE,

(ii) the area of the trapezium ABCD.

(iii) the length of BC. [8]

(b) Given that

B = {b: 0 < 2b – 3 < 6, b is an integer},

write down the elements of B. [2]

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5 (a) Solve the equation $3x^2 - 5x - 9 = 0$, giving your answers correct to two significant figures. [5]

(b) Given that y varies inversely as the square of x and that $y = 3$ when $x = 2$,

(i) find the equation connecting x and y and show that it reduces to $x = \sqrt{\frac{12}{y}}$,

(ii) calculate the value of x when $y = 5\frac{1}{3}$. [5]

6 (a) Given that $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 4 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 2 \\ 0 & 5 \end{pmatrix}$, find

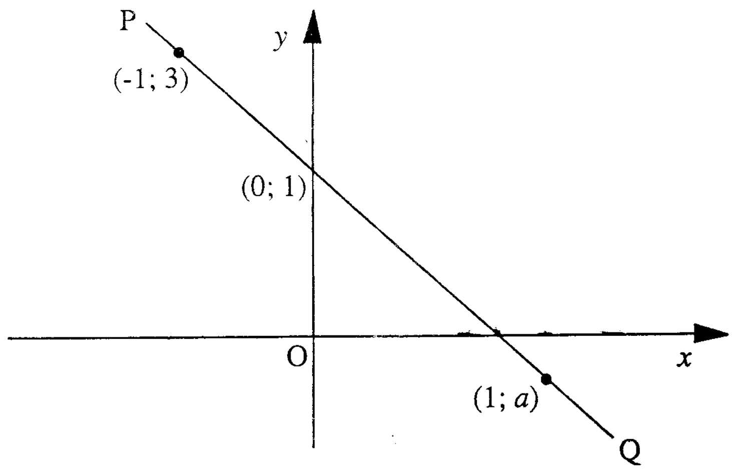
(i) the determinant of $\mathbf{A}$,

(ii) $\mathbf{A}^{-1}$,

(iii) $\mathbf{B}^2 - \mathbf{A}$.

[6]

(b)



In the diagram , PQ is a straight line passing through the points $(-1; 3)$, $(0; 1)$ and $(1; a)$.

(i) Find the numerical value of the gradient of PQ.

(ii) Calculate the value of a.

(iii) Write down the equation of the line PQ. [5]

Section B [36 marks]

*Answer any **three** questions from this section. Each question carries 12 marks.*

7 **Answer the whole of this question on a sheet of plain paper.**

*Use ruler and compasses only for all constructions and show clearly all construction arcs and lines on a **single** diagram.*

P, Q, R and T are four villages. P is due north of T. The bearing of Q from P is 240°. P, Q and R are in a straight line in that order. PR = 10 km, QR = 2 km and $P\hat{R}T = 45°$.

(a) Using a scale of 1 cm to 2 km construct a diagram to show the relative positions of P, Q, R and T. [5]

(b) Construct the locus of points

(i) equidistant from P and R,

(ii) 5 km from T. [3]

(c) A school is to be built such that it is equidistant from P and R and 10 km from T, mark and label clearly S_1 and S_2 the two possible positions of the school. [2]

(d) Use the diagram to find

(i) the actual distance of Q from T,

(ii) the bearing of R from T. [2]

8 ***Answer the whole of this question on a sheet of graph paper.***

The following is an incomplete table of values for the function
$y = x^3 - 6x^2 + 3x + 10$.

x	$-1,5$	-1	0	1	2	3	4	5	$5,5$
y	$-11,4$	0	10	p	0	-8	-10	0	$11,4$

(a) Calculate the value of p. [1]

(b) Using a scale of 2 cm to represent 1 unit on the x-axis and 2 cm to represent 5 units on the y-axis, draw the graph of $y = x^3 - 6x^2 + 3x + 10$ for $-1,5 \le x \le 5,5$. [4]

(c) Use the graph to write down

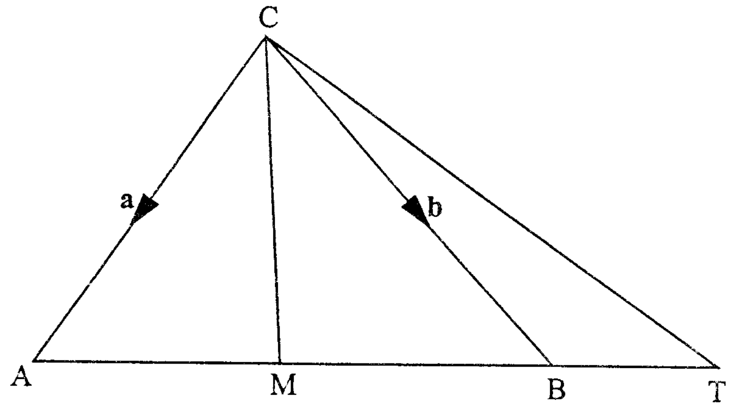
(i) the coordinates of the maximum turning point of the curve,

(ii) the three solutions of the equations $x^3 - 6x^2 + 3x + 10 = 0$. [4]

(d) Find the gradient of the curve when $x = 4$. [2]

(e) On the graph shade the area bounded by the curve $y = x^3 - 6x^2 + 3x + 10$ and the lines $y = 0$, $x = -\frac{1}{2}$ and $x = 1$. [1]

9 **(a)**



In the diagram, ABC is a triangle and M is the mid-point of AB. ABT is a straight line and $\frac{MB}{MT} = \frac{2}{3}$.

$\overrightarrow{CA} = \mathbf{a}$ and $\overrightarrow{CB} = \mathbf{b}$.

Express the following vectors in terms of **a** and /or **b**

(i) $\overrightarrow{AB}$,

(ii) $\overrightarrow{MT}$,

(iii) $\overrightarrow{CT}$. [5]

(b) Two circles have areas of $6\frac{1}{4}$ m^2 and $2\frac{1}{2}$ m^2 respectively.

(i) Express the area of the smaller circles as a percentage of the area of the large circle.

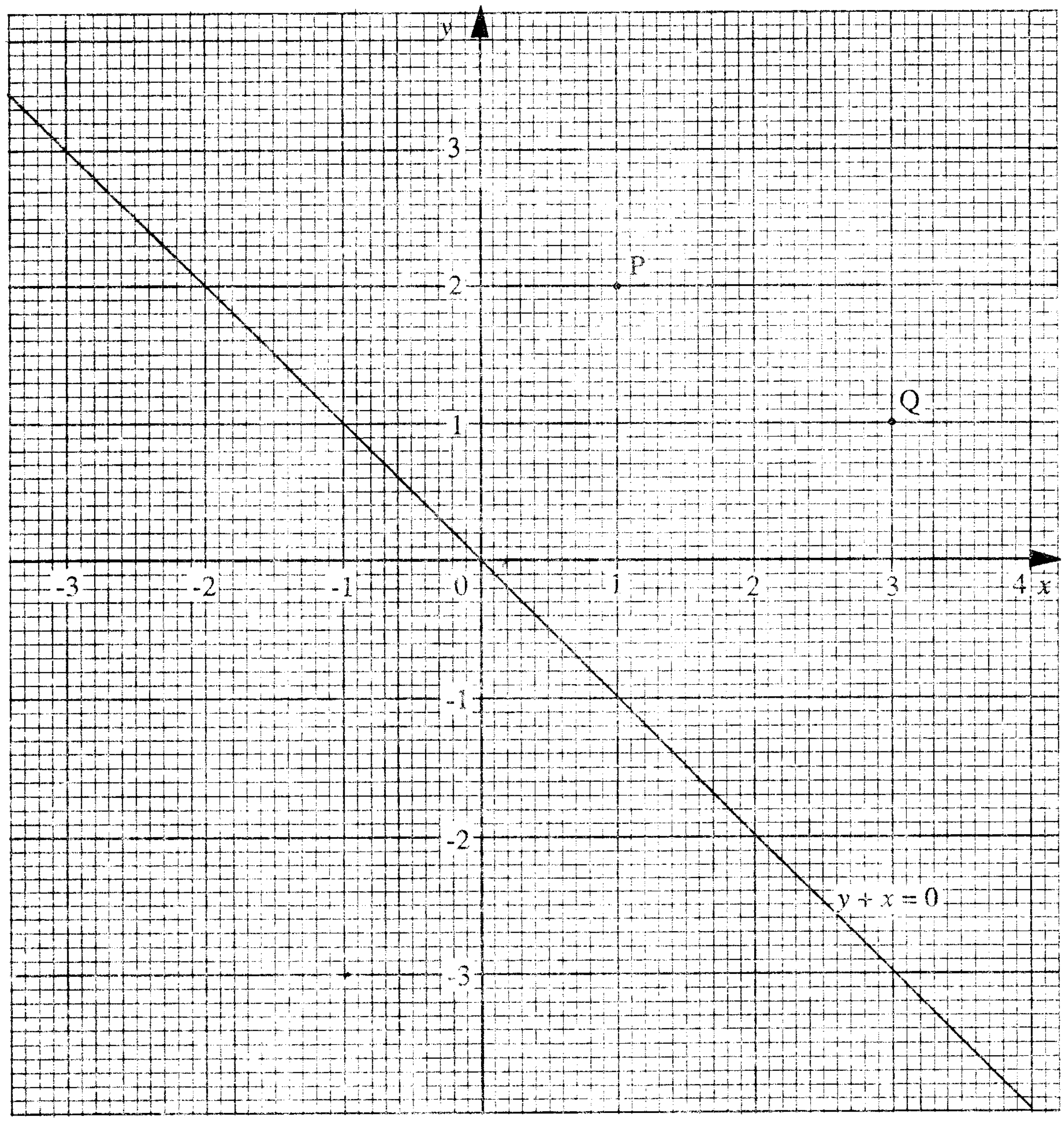
(ii) Calculate the ratio $\dfrac{\text{radius of the larger circle}}{\text{radius of the smaller circle}}$

leaving your answer in surd form.

(iii) (In this part **take** π **to be** $\frac{22}{7}$).

Calculate the radius of the larger circle. [7]

10 (a) *Do not copy this diagram.*



In the diagram, P is the point (1; 2) and Q is the point (3; 1). Write down the coordinates of

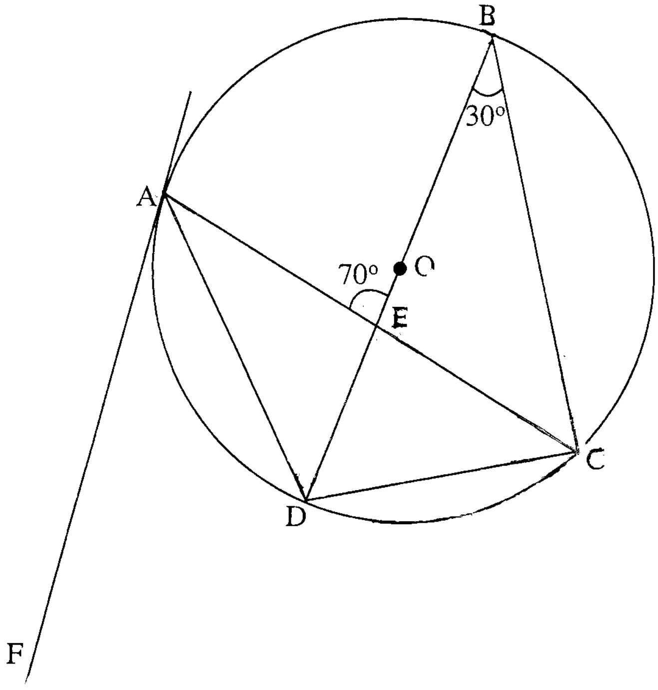
(i) R, the reflection of P in the line $y + x = 0$.

(ii) S, the image of Q under a clockwise rotation of 90° about the origin.

(iii) T, the image of P under a translation $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

(iv) U, the image of Q under a shear of factor $-1\frac{1}{2}$ with the x-axis invariant. [7]

(b)



In the diagram, A,B,C and D are points on a circle centre O. BD is a diameter and AF is the tangent to the circle at A. $C\hat{B}D = 30°$ and AC and BD intersect at E such that $A\hat{E}B = 70°$.

Find

(i) $C\hat{A}D$,

(ii) $A\hat{D}B$,

(iii) $A\hat{C}D$,

(iv) $C\hat{A}F$. [5]

11 Answer the whole of this question on a sheet of graph paper.

Mark (x)	$0< x \le 10$	$10< x \le 20$	$20< x \le 30$	$30< x \le 40$	$40< x \le 60$	$60 < x \le 70$	$70 < x \le 80$	$80 < x \le 90$	$90 < x \le 100$
Frequency	8	15	p	29	83	20	10	10	5
Cumulative Frequency	8	23	43	72	155	180	q	198	200

Above is a cumulative frequency table for the distribution of marks obtained by 200 students in a Geography test.

(a) Find the value of p and the value of q. [2]

(b) Using a scale of 2 cm to represent 20 marks on the horizontal axis and 2 cm to represent 20 students on the vertical axis, draw the cumulative frequency curve for the distribution. [4]

(c) Use the graph to estimate

(i) the median mark,

(ii) the number of students who scored 75 marks or more. [3]

(d) Calculate an estimate of the mean mark for the top 45 students. [3]

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12 (a) In 2004, a motorist filled up his fuel tank with 45 litres of fuel bought at \$3 450 per litre.

(i) Calculate the amount he paid for the fuel.

(ii) If the price of fuel per litre included 15% Value Added Tax (VAT), calculate the price of fuel per litre without the VAT.

(iii) The motorist then used 12,5 litres in travelling a distance of 196 kilometres. Calculate the rate of fuel consumption of the car in kilometres per litre. [6]

(b) Study the patterns below and answer the questions that follow.

	Column 1	Column 2	Column 3
Row 1	2	2	4
2	3	7	10
3	4	14	18
4	5	23	28
5	6	34	40
6			
•	•	•	•
•	•	•	•
•	•	•	•
17		u	340
•	•	•	•
•	•	•	•
t			w
•			•
•			•
•			•

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(i) Write down the sixth row of the pattern.

(ii) Write down the value of u.

(iii) Express w in terms of t. [6]